Supplementary Materials for 'The instability of the two cultural attractors found within the Renaissance'
immediate

Second rater validation

To further assess the plausibility of automated gaze estimation across corpora, an independent rater evaluated a stratified random sample of 100 portraits drawn from Renaissance secular (n = 40), post-Renaissance secular (n = 20), Renaissance religious (n = 20), and ukiyo-e datasets (n = 20). The rater judged whether the OpenFace-derived horizontal gaze direction appeared plausible relative to the painted eye orientation (yes/no/unsure). Vertical gaze was not evaluated. This additional validation step allows us to evaluate whether detection plausibility differed systematically across artistic traditions. The independent rater provided "unsure" responses only for ukiyo-e stimuli. The primary reason for this uncertainty was that some ukiyo-e portraits depicted stylized cross-eyed expressions (*neko-me*, or "cat eyes"), making it difficult to judge whether the estimated gaze angle accurately reflected the intended artistic portrayal (n = 3). For these stimuli, disagreements were resolved through discussion between the rater and the author.

Renaissance gaze analyzed with cone of direct gaze

Coding of direct and averted gaze

To classify each portrait into *direct* versus *averted gaze*, we implemented a continuous probabilistic rule that accounts for both **head orientation** (`headYaw`) and **eye orientation** (`gazeYaw`). `headYaw` was coded in degrees such that 0° corresponds to a frontal head orientation and positive values indicate rightward rotation. `gazeYaw` was coded such that 0° corresponds to direct frontal gaze and positive values indicate leftward deviation.

Because the perceptual "cone of direct gaze" narrows as the head rotates away from the frontal position, we defined a head-dependent tolerance parameter $\tau(h)$. Specifically, when the head was frontal (0°), gaze was considered "direct" within $\pm 7.5°$; when the head was rotated 30°, the tolerance narrowed to $\pm 3°$. For intermediate angles between 0° and 30°, we linearly interpolated the tolerance; values beyond 30° were fixed at $\pm 3°$.

To model the transition smoothly rather than as a hard cutoff, we converted this rule into a cumulative normal distribution: $p_{\text{direct}} = \Phi\left(\frac{\tau(|h|)-|g|}{\sigma}\right)$,

where $|h|$ is the absolute headYaw, $|g|$ is the absolute gazeYaw, $\tau(|h|)$ is the tolerance at that head orientation, σ is a slope parameter (set to 2° in the main analyses), and Φ is the standard normal cumulative distribution function.

This formulation produces a probability of direct gaze that decreases gradually as gaze deviates from the head-dependent tolerance. For categorical analyses, we coded trials as "direct" when $p_{\text{direct}} \geq 0.5$, and as "averted" otherwise. In robustness checks, we also sampled classifications probabilistically from p_{direct}.

Direct gaze during the Renaissance

To examine diachronic changes in gaze orientation during the Renaissance, we estimated a logistic regression model predicting whether a portrait depicted a direct gaze (1) versus an averted gaze (0) from the date of production (restricted to 1400–1600). The model revealed a significant negative intercept ($\beta = -7.71$, $SE = 3.62$, $p = .033$), reflecting the overall rarity of direct gaze at the beginning of the period. The coefficient for date was positive ($\beta = 0.004$, $SE = 0.002$, $p = .082$), indicating that portraits produced later in the Renaissance were somewhat more likely to be depicted with direct gaze. Although this effect did not reach conventional levels of statistical significance, the trend suggests a gradual increase in direct gaze between 1400 and 1600.

Renaissance secular paintings and religious paintings using cone of direct gaze

Evolution of gaze within the two conditions

To assess whether diachronic changes in gaze orientation differed across stylistic contexts, we fitted a logistic regression model predicting the likelihood of direct gaze from date, Condition, and their interaction (1400–1600. The analysis revealed a significant negative effect of date in the religious paintings ($\beta = -0.0097$, $SE = 0.0047$, $p = .038$), indicating that direct gaze became less likely over time in religious paintings. By contrast, a significant positive interaction emerged between date and the secular paintings ($\beta = 0.0138$, $SE = 0.0052$, $p = .008$). This pattern suggests that while religious portraits showed a slight decline in direct gaze, secular portraits exhibited the opposite tendency, with a gradual increase in the likelihood of direct gaze as the period progressed. The main effect of the secular portraits was also significant ($\beta = -21.75$, $SE = 7.82$, $p = .005$), reflecting the lower baseline probability of direct gaze at the start of the Renaissance period.

Popularity

To test whether direct gaze predicted the cultural prominence of portraits, we fitted a linear mixed-effects model with the log-transformed number of web search hits as the dependent variable, gaze category (0 = averted, 1 = direct) and sitter fame (log-transformed number of Wikipedia language editions) as fixed effects, and author as a random intercept.

The model revealed a strong positive effect of sitter fame ($\beta = 0.27$, $SE = 0.04$, $t = 6.53$, $p < .001$), indicating that portraits of more famous sitters were consistently associated with greater cultural prominence. By contrast, gaze orientation had no reliable effect ($\beta = -0.02$, $SE = 0.11$, $t = -0.18$, $p = .858$). Thus, while sitter fame strongly predicted the visibility of a portrait in contemporary search metrics, there was no evidence that direct gaze independently contributed to cultural prominence.

Using cone of direct gaze for the ukiyo-e paintings

We next examined whether the prevalence of direct gaze changed over time in the ukiyo-e sample. A logistic regression was fitted with gaze category (0 = averted, 1 = direct) predicted by year of production. The model revealed no significant association between year and the likelihood of direct gaze ($\beta = -0.026$, $SE = 0.022$, $p = .243$). The intercept was also non-significant ($\beta = 46.02$, $SE = 41.60$, $p = .269$). In sum, the analysis provided no evidence for a diachronic increase (or decrease) in direct gaze in ukiyo-e portraits across the period examined.

Tables of regression models in the main article

Predictor	β	*SE*	*95%CI*	*t*	*df*	*p*	βStd
Intercept	-15.74	0.90	[-17.494, -13.983]	-17.57	5.43	<.001	
date	0.06	0.01	[0.032, 0.091]	4.13	144.26	<.001	0.26

Table 1. Fixed Effects Estimates from

Group	Variance	*SD*
School	31.70	5.63
Artist	1.63	1.28
Residual	51.68	7.19

Table 2. Random Effects Variance Components

Predictor	β	*SE*	*95%CI*	*t*	*df*	*p*	βStd
Intercept	14.55	0.76	[13.055, 16.048]	19.06	76.42	<.001	
sdate	-0.04	0.02	[-0.069, 0]	-1.98	126.27	0.049	-0.15

Table 3. Fixed Effects Estimates

===== LATEX: Fixed Effects Table =====
===== LATEX: Coefficient Table =====
===== LATEX: Odds Ratio Table =====

Group	Variance	SD
School	25.55	5.05
Artist	0.00	0.00
Residual	65.68	8.10

Table 4. Random Effects Variance Components

Predictor	β	*SE*	95%*CI*	*t*	*df*	*p*	βStd
(Intercept)	4.152	0.346	[3.473, 4.831]	11.990	3.420	<.001	
Gaze (centered)	-0.004	0.010	[-0.024, 0.015]	-0.440	254.530	0.663	-0.019
Log Fame (centered)	0.492	0.076	[0.342, 0.642]	6.440	259.260	<.001	0.293
Date (centered)	-0.002	0.004	[-0.01, 0.007]	-0.360	87.100	0.72	-0.027

Table 5. Fixed Effects Estimates (Renaissance, 1400–1600)

Group	Variance	*SD*	*ICC*
School	0.518	0.720	0.158
Artist	1.117	1.057	0.342
Residual	1.634	1.278	

Table 6. Random Effects Variance Components (Renaissance, 1400–1600)

Predictor	β	*SE*	95%*CI*	*t*	*df*	*p*	βStd
(Intercept)	11.505	1.676	[8.219, 14.791]	6.860	100.740	<.001	
Date (centered)	-0.033	0.010	[-0.054, -0.013]	-3.210	157.750	0.002	-0.182

Table 7. Fixed Effects Estimates (Renaissance, Three-Quarter Pose)

Group	Variance	*SD*	*ICC*
School	2.267	1.506	0.040
Author	15.343	3.917	0.271
Residual	39.033	6.248	

Table 8. Random Effects Variance Components (Renaissance, Three-Quarter Pose)

Predictor	β	*SE*	95%*CI*	*t*	*df*	*p*	β *std*
(Intercept)	16.238	0.505	[15.248, 17.227]	32.160	7.410	<.001	
Date (centered)	0.020	0.005	[0.01, 0.03]	3.950	133.190	<.001	0.177

Table 9. Fixed Effects Estimates (Renaissance, Three-Quarter Pose)

Group	Variance	*SD*	*ICC*
School	1.693	1.301	0.033
Author	2.967	1.722	0.058
Residual	46.294	6.804	

Table 10. Random Effects Variance Components (Renaissance, Three-Quarter Pose)

Predictor	β	*SE*	95%*CI*	*t*	*df*	*p*	β*Std*
(Intercept)	12.416	0.582	[11.275, 13.558]	21.320	8.610	<.001	
Date (centered)	0.015	0.005	[0.004, 0.025]	2.680	202.950	0.008	0.099

Table 11. Fixed Effects Estimates (Post-Renaissance, 1600+)

Group	Variance	*SD*	*ICC*
School	2.877	1.696	0.036
Author	7.126	2.669	0.090
Residual	69.208	8.319	

Table 12. Random Effects Variance Components (Post-Renaissance, 1600+)

Predictor	β	*SE*	95%*CI*	*t*	*df*	*p*	β*Std*
Intercept	4.123	0.359	[3.419, 4.826]	11.490	3.830	<.001	
Head Direction: Right Cheek	0.042	0.199	[-0.348, 0.432]	0.210	256.250	0.832	
Head Direction: Straight	0.428	0.312	[-0.183, 1.039]	1.370	244.880	0.171	
Log Fame	0.601	0.093	[0.419, 0.784]	6.450	258.250	<.001	0.601
Date	-0.065	0.154	[-0.366, 0.237]	-0.420	85.850	0.676	-0.065

Table 13. Fixed Effects Estimates: Head Direction Model (Renaissance, <1600)

Group	Variance	*SD*	*ICC*
School	0.537	0.733	0.163
Author	1.130	1.063	0.343
Residual	1.625	1.275	

Table 14. Random Effects Variance Components (Head Direction Model, <1600)

Outcome	Predictor	Coefficient	*SE*	*z*	*p*
Right Cheek vs Left Cheek	Intercept	0.001	0.065	0.010	0.989
Right Cheek vs Left Cheek	Date (standardized)	-0.025	0.065	-0.380	0.701
Straight vs Left Cheek	Intercept	-0.819	0.084	-9.760	<.001
Straight vs Left Cheek	Date (standardized)	0.258	0.085	3.030	0.002

Table 15. Multinomial Logistic Regression: Log-Odds Coefficients (Post Renaissance, >1600)

Outcome	Predictor	*OR*	95%*CI*
Right Cheek vs Left Cheek	Intercept	1.001	[0.881, 1.137]
Right Cheek vs Left Cheek	Date (standardized)	0.975	[0.859, 1.108]
Straight vs Left Cheek	Intercept	0.441	[0.374, 0.52]
Straight vs Left Cheek	Date (standardized)	1.294	[1.095, 1.528]

Table 16. Multinomial Logistic Regression: Odds Ratios with 95% CI (Post Renaissance, >1600)

Predictor		SE	95%*CI*	*t*	*df*	*p*	*std*
(Intercept)	3.838	0.401	[3.052, 4.625]	9.570	15.210	<.001	
Head Direction: Right Cheek	0.006	0.267	[-0.518, 0.53]	0.020	253.010	0.982	
Head Direction: Straight	-0.088	0.363	[-0.8, 0.624]	-0.240	251.830	0.808	
Log Fame (z-score)	0.656	0.127	[0.407, 0.904]	5.170	258.950	<.001	0.656
Date (z-score)	0.313	0.163	[-0.006, 0.632]	1.920	79.500	0.058	0.313

Table 17. Fixed Effects Estimates: Head Direction Model (Post Renaissance, >1600)

Group	Variance	*SD*	*ICC*
School	1.506	1.227	0.275
Author	0.458	0.677	0.084
Residual	3.509	1.873	

Table 18. Random Effects Variance Components (Head Direction Model, >1600)

Predictor	β	*SE*	95%*CI*	*t*	*p*	βstd
Intercept	82.397	125.763	[-168.844, 333.638]	0.660	0.515	
Year	-0.031	0.068	[-0.166, 0.104]	-0.450	0.651	-0.055

Table 19. Fixed Effects: Ukiyo-e Gaze Deviation $\sim$ Year

Outcome	Predictor	Coefficient	*SE*	*z*	*p*
Right Cheek vs Left Cheek	Intercept	-0.979	0.283	-3.450	<.001
Right Cheek vs Left Cheek	Year (standardized)	0.393	0.298	1.320	0.187
Straight vs Left Cheek	Intercept	-8.926	9.709	-0.920	0.358
Straight vs Left Cheek	Year (standardized)	3.831	4.875	0.790	0.432

Table 20. Multinomial Logistic Regression: Log-Odds Coefficients (Ukiyo-e)

Outcome	Predictor	*OR*	95%*CI*
Right Cheek vs Left Cheek	Intercept	0.376	[0.216, 0.655]
Right Cheek vs Left Cheek	Year (standardized)	1.482	[0.826, 2.658]
Straight vs Left Cheek	Intercept	0.000	[0, 24429.893]
Straight vs Left Cheek	Year (standardized)	46.097	[0.003, 650474.972]

Table 21. Multinomial Logistic Regression: Odds Ratios with 95% CI (Ukiyo-e)